Finding the Mistake in Sums of Radicals

Testing whether two roots may be added, combining like radicals, and simplifying before combining

Directions

- Two radical terms may be added only when the part *under* the radical is already the same. Then you add the counts in front and leave the radical alone.
- A radical sign never distributes across a plus sign. When you suspect it has been treated as if it does, test the two sides with numbers.
- Leave answers in radical form unless a problem asks for a decimal.

1. Test the two sides of a tempting shortcut.

(a) $\sqrt{9} + \sqrt{16} =$ _________

(b) $\sqrt{9 + 16} =$ _________

(c) Write $<$, $>$, or $=$ between your two answers: (a) _________ (b)

2. Priya writes $\sqrt{36} + \sqrt{64} = \sqrt{36 + 64} = 10$.

(a) Work out the correct value of $\sqrt{36} + \sqrt{64}$. Answer: _________

(b) Explain what Priya did wrong, and state the rule she broke.

3. Combine into a single radical term: $3\sqrt{2} + 5\sqrt{2}$.

Answer: ____________

4. Combine into a single radical term: $7\sqrt{5} - 2\sqrt{5}$.

Answer: ____________

5. These roots are not exact, so test the shortcut with rounded decimals.

(a) To the nearest hundredth, $\sqrt{2} + \sqrt{3} \approx$ ________

(b) To the nearest hundredth, $\sqrt{5} \approx$ ________

(c) Write $<$, $>$, or $=$ between your two answers: (a) ________ (b)

6. Simplify $\sqrt{8} + \sqrt{18}$ into a single radical term. Rewrite each root first, using its largest perfect-square factor.

Answer: ________________

7. Marco writes $4\sqrt{3} + 2\sqrt{3} = 6\sqrt{6}$.

(a) Work out the correct value of $4\sqrt{3} + 2\sqrt{3}$. Answer: ________

(b) Explain what Marco did wrong. Which part of the term is allowed to change, and which part is not?

8. Challenge. Consider $\sqrt{50} + \sqrt{32} - \sqrt{8}$.

(a) Simplify it into a single radical term. Answer: _________

(b) Justify why the three terms could be combined at all, even though the three numbers under the radicals started out different.